



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 2

Central Bank, Commercial Banks and Investment Banks



STUDY NOTES • EDITION 1.0 • WEEK 2

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How to use these notes



These notes cover the whole of Week 2: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A covers money itself and the central bank that manages it. Part B takes commercial banking apart, from the balance sheet to credit creation. Part C turns to investment banks, which do something entirely different with capital. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

Contents



GETTING ORIENTED

How to use these notes	02
Module overview, objectives and topic map	03
Key terms at a glance	04

A Money and the central bank

1	What money is, and what it does
2	Money markets and the money supply
3	Monetary policy and financial stability

B Commercial banking

4	What a commercial bank actually does
5	Credit creation and the money multiplier
6	Off-balance sheet activity and bank risk

C Investment banking

7	What an investment bank is
8	Underwriting and the IPO process
9	Mergers, acquisitions and the deal cycle

CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



This module examines the three institutions at the centre of any financial system. It begins with money itself — what it does, how it evolved, and how its supply is measured and controlled — then moves to commercial banks, which turn deposits into credit, and closes with investment banks, which raise capital and advise on transactions but never take a deposit.

One thread runs through all nine topics: **every institution here transforms something**. The central bank transforms policy into liquidity, the commercial bank transforms short deposits into long loans, and the investment bank transforms a company's prospects into a price the market will pay.

Learning objectives

By the end of this module, you will be able to:

- 1 **Explain** the three functions of money, trace its evolution, and describe how the money supply is measured.
- 2 **Describe** the tools of monetary policy, distinguish base from multiplier effects, and explain the lender-of-last-resort role.
- 3 **Explain** how commercial banks create credit, apply the money multiplier, and describe how bank risk is screened and controlled.
- 4 **Distinguish** investment from commercial banking, and explain underwriting, IPO pricing and the M&A process.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	What money is, and what it does	A • Money	L5 V1	LO 1
2	Money markets and the money supply	A • Money	L5 V1	LO 1
3	Monetary policy and financial stability	A • Money	L5 V2	LO 2
4	What a commercial bank actually does	B • Commercial	L6 V1	LO 3
5	Credit creation and the money multiplier	B • Commercial	L6 V2	LO 3
6	Off-balance sheet activity and bank risk	B • Commercial	L6 V1	LO 3
7	What an investment bank is	C • Investment	L7 V1	LO 4
8	Underwriting and the IPO process	C • Investment	L7 V2	LO 4
9	Mergers, acquisitions and the deal cycle	C • Investment	L7 V2	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Double coincidence of wants

The barter problem where two parties must each want exactly what the other has.

M1, M2, M3

Graded measures of money supply, each adding a less liquid component to the one before.

Monetary base

Currency in circulation plus all bank reserves held at the central bank.

Reserve requirement

The minimum fraction of every deposit a bank must retain as liquid reserves.

Open market operations

Central bank purchase or sale of government securities to inject or drain reserves.

Repo rate

India's policy rate, signalling the cost of money to the banking system.

Lender of last resort

Emergency central bank liquidity to institutions that are illiquid but not insolvent.

Asset transformation

Borrowing short and lending long, converting illiquid risky assets into safer liquid ones.

Five Cs of credit

Character, capacity, collateral, conditions and capital — the loan screening framework.

Off-balance sheet activity

Commitments and contingencies generating fee income without appearing as assets or liabilities.

Underwriting

A bank buying an entire securities issue at a fixed price and reselling it to investors.

Best efforts

An arrangement where the bank sells what it can without committing to buy the issue.

Junk bonds

High-yield debt with minimal collateral, letting buyers acquire targets larger than themselves.

Fiat money

Currency declared legal tender by government, requiring trust in the issuing authority.

Velocity of money

The rate at which money circulates through the economy.

Money multiplier

How far the banking system expands the base through lending. $MM = 1 \div \text{reserve ratio}$.

Excess reserves

The portion of a deposit above the required reserve — the lending fuel.

Discount rate

The rate at which banks borrow from the central bank.

Inflation targeting

A public commitment to a precise inflation target — the RBI's is 4%, within a 2–6% band.

Moral hazard

The incentive to take reckless risk when rescue is expected.

Net interest margin

The spread between interest earned on loans and interest paid on deposits.

Capital adequacy ratio

The regulatory minimum equity a bank must hold to absorb losses, under Basel III.

Value at risk

Maximum potential loss over a horizon under normal conditions at a given confidence level.

Underpricing

Setting an issue price below what the market would bear — capital the issuer forgoes.

Tender offer

A public bid to buy shares at a premium, made directly to shareholders over management's objection.

PART A • MONEY AND THE CENTRAL BANK

Money, and the institution that manages it

1 What Money Is, and What It Does

PART A • MONEY AND THE CENTRAL BANK • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Money exists to solve one problem — the double coincidence of wants — and everything the central bank does follows from managing what money then becomes.

Why this matters

Every institution in this week manages money in some form. If the three functions are not fixed in your mind, the monetary policy transmission in Topic 3 and the credit creation in Topic 5 will look like unrelated machinery.

The problem money solves

A farmer has fifty kilogrammes of wheat and needs shoes. The cobbler needs cloth. The cloth merchant needs wheat — but only twenty kilogrammes. Nobody can agree, so nobody trades, and everyone goes home empty-handed. This is the **double coincidence of wants**, and it paralysed economies for centuries. Introduce a coin everyone agrees has value and the farmer sells to anyone, takes the coins, and buys the shoes.

The three functions

Medium of exchange. Money buys and sells goods and services, eliminating barter and reducing transaction costs. **Unit of account.** It provides a common measure of value, simplifying price comparison and economic calculation. **Store of value.** It preserves purchasing power over time, allowing households and firms to save.

Five economic benefits

Adopting money reduces transaction costs and improves efficiency; eliminates the double coincidence of wants; encourages specialisation and the division of labour; acts as the most liquid asset in the economy; and facilitates the smooth functioning of markets and economic activity.

How money evolved

Commodity money — gold and silver — carried intrinsic value and was widely accepted, but was difficult to transport. **Paper currency**, or fiat money, is declared legal tender by government: lightweight and convenient, but requiring trust in the issuing authority, which in India is the Reserve Bank of India. **Banking innovations** brought cheques, reducing the need to carry cash. **Digital innovations** continue the line — UPI is rapidly replacing cash for everyday transactions across India.

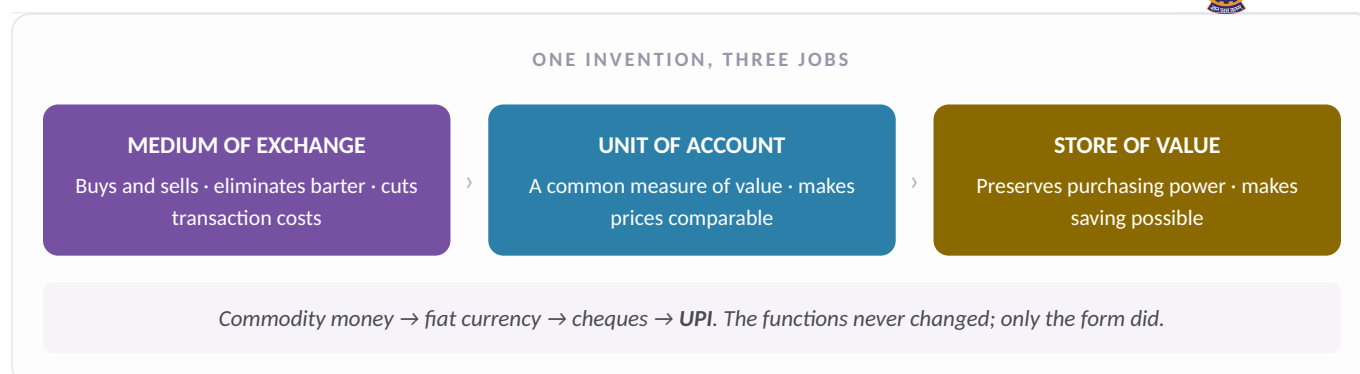


Figure 1.1 The three functions of money, and the evolution of the form that performs them.

KEY IDEA

Fiat money has no intrinsic value at all. It works purely because everyone trusts the issuing authority — which is precisely why the credibility of a central bank is an economic asset, not a constitutional nicety.

EXAM POINTER

Name all three functions with a one-line explanation each, then give at least three of the five benefits. The benefit most often forgotten is that money *encourages specialisation* — without it, everyone must produce what they personally need.

QUICK RECAP • TOPIC 1

- Money solves the double coincidence of wants, which otherwise blocks trade entirely.
- Three functions: medium of exchange, unit of account, store of value.
- Five benefits, including specialisation and being the most liquid asset in the economy.
- Evolution: commodity money, fiat currency, cheques, and now digital payment through UPI.
- Fiat money works only on trust in the issuing authority — in India, the RBI.

2 Money Markets and the Money Supply

PART A • MONEY AND THE CENTRAL BANK • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Short-term liquidity trades through six instruments, and the central bank tracks the resulting money supply in three graded layers.

Why this matters

M1, M2 and M3 are the quantities monetary policy actually targets. The instruments below are also where Week 3 begins its treatment of securities, so the vocabulary pays twice.

Money market securities

Low-risk, highly liquid debt instruments maturing in one year or less. Together they form the core of the money market, providing safe and liquid options for short-term borrowing and investing.

Instrument	What it is	Maturity	Secondary market
Treasury bills	Issued by government	91 days to 1 year	Highly active
Commercial paper	Unsecured promissory notes from large firms and financial institutions	1 day to 9 months	Moderate
Negotiable CDs	Issued by depository institutions	Up to 1 year	Low
Bankers' acceptances	Time drafts guaranteed by banks, used in international trade	Up to 6 months	Highly active
Repurchase agreements	Securities sold with an agreement to repurchase later	Up to 1 year	None
Federal funds	Very short-term loans between depository institutions	1 day to 1 week	None

India note: the lecture uses US instruments. The Indian equivalents are treasury bills issued by the Government of India through the RBI, and the call money market in place of federal funds.

Three measures of money supply

Each measure adds a less liquid component to the one before it, so the progression from M1 to M3 captures increasing breadth of liquidity. The RBI uses these aggregates for policy formulation and economic analysis.

The quantity theory of money

How does the money supply translate into economic activity? Through the identity $MS \times VM = GDP = RO \times PL$, where MS is the money supply, VM the velocity of money (the rate at which it circulates), GDP nominal output, RO real output and PL the price level. In words: total spending in the economy equals real output multiplied by the price level.

Two schools read the transmission differently. **Monetarists** argue money supply changes affect GDP directly — more money means more nominal output, and if real output is near capacity, prices rise. **Keynesians** propose an indirect channel: money supply moves interest rates, which move investment and consumption, which move GDP.

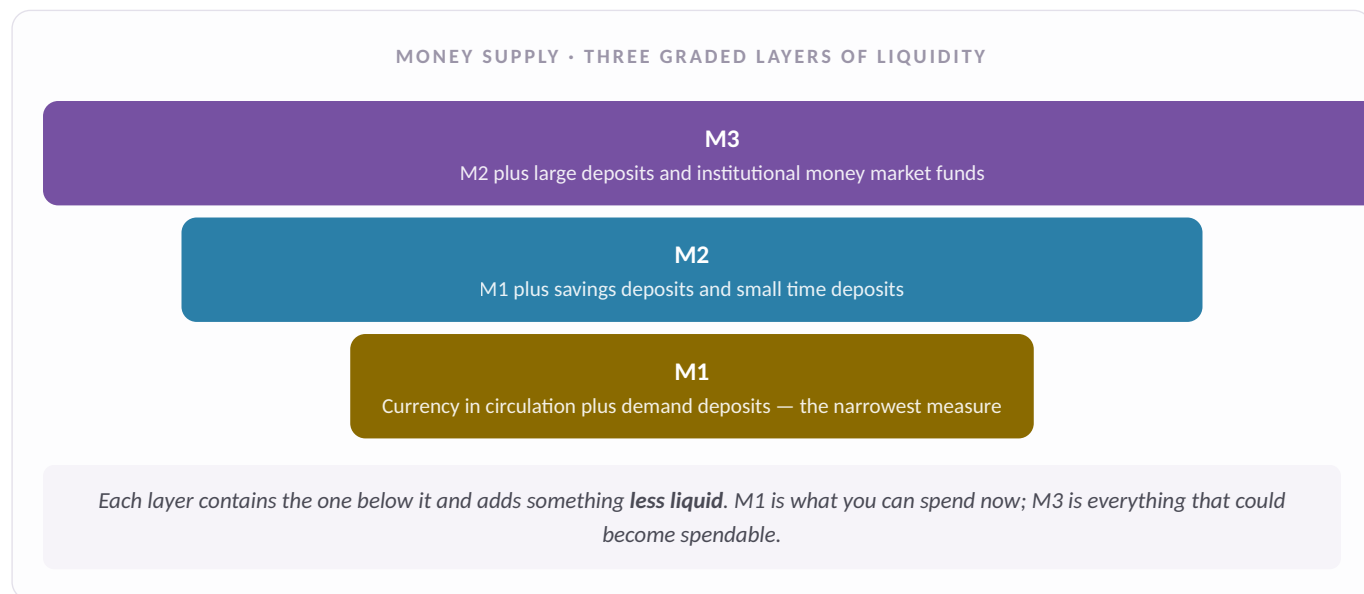


Figure 2.1 The money supply measures. Read outwards from M1: each adds a less liquid component.

COMMON PITFALL

Treating M1, M2 and M3 as alternatives. They are *nested* — M3 contains M2, which contains M1. A question asking what distinguishes them wants the additional component at each step, not three separate definitions.

EXAM POINTER

For the quantity theory, write the identity in full and define all five terms before interpreting it. Then contrast monetarist and Keynesian transmission in one sentence each — direct versus via interest rates. That contrast is the examinable point, not the algebra.

QUICK RECAP • TOPIC 2

- Money market instruments mature in a year or less: T-bills, commercial paper, CDs, acceptances, repos, interbank loans.
- They are low-risk, highly liquid and low-yielding by design.
- M1 is currency plus demand deposits; M2 adds savings and small time deposits; M3 adds large and institutional funds.
- Quantity theory: $MS \times VM = GDP = RO \times PL$.
- Monetarists see a direct effect on output; Keynesians see it working through interest rates.

3 Monetary Policy and Financial Stability

PART A • MONEY AND THE CENTRAL BANK • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Three tools, two directions, and one persistent tension between growth, prices and employment.

Why this matters

This is the topic that connects a decision in Mumbai to a household's loan instalment. It also supplies the reserve requirement, which Topic 5 turns into the money multiplier.

Three objectives in tension

The central bank pursues **economic growth** (sustainable and non-inflationary), **price stability** (rapid price rises erode purchasing power and raise uncertainty), and **full employment**. These can conflict: stimulating growth may fuel inflation, and controlling inflation may slow growth. Managing that trade-off is the essence of monetary policy.

Three tools

Tool	What it is	How it works	Acts through
Open market operations	Buying or selling government securities	Buying injects money; selling absorbs it. The most frequently used tool	The monetary base
Discount rate	The rate at which banks borrow from the central bank	Raising it means banks borrow less; lowering it, more. Signals policy direction	The monetary base
Reserve requirements	The minimum fraction banks must hold	A higher requirement means less lending capacity; a lower one means more credit creation	The money multiplier

The key distinction: reserve requirements change what the system can multiply. Open market operations and the discount rate change what there is to multiply. The *monetary base* is total currency in circulation plus all bank reserves held at the central bank — only the central bank can change it.

Two directions

Expansionary policy is the accelerator: money supply up, interest rates down, aggregate demand up. Used when growth is weak — the RBI cut the repo rate from 5.15% to 4% during COVID-19 to inject liquidity. **Contractionary policy** is the brake: money supply down, rates up, demand controlled. Used when the economy overheats — the US Federal Reserve raised rates aggressively through 2022 and 2023 to tame forty-year-high inflation. Both directions use the same three instruments; only the direction of use differs.

Inflation targeting and financial stability

Under **explicit inflation targeting** the central bank declares a precise target — the RBI's is 4%, within a band of 2% to 6% — and must publicly explain sustained deviation. It works because a credible target gets built into wage negotiations, pricing decisions and long-term contracts. Belief itself reduces volatility.

The third dimension is **financial stability**. As *lender of last resort*, the central bank provides emergency funding when an institution faces a liquidity crisis. The critical distinction is that the institution must be **illiquid, not insolvent**: this is a bridge, not a bailout. In 2008 the US Federal Reserve extended emergency credit, purchased distressed assets and injected liquidity on an unprecedented scale, actions widely credited with preventing a second Great Depression. Support must nonetheless be conditional, because unconditional rescue encourages reckless risk-taking — *moral hazard*.

SIX ACTIONS • THREE EXPAND THE MONEY SUPPLY, THREE CONTRACT IT

Action	Effect on banks	Base or multiplier	Money supply	Stance
Discount rate up	Borrowing costs rise	Base shrinks	Decreases	Contractionary
Discount rate down	Borrowing is cheaper	Base expands	Increases	Expansionary
Reserve requirement up	Less available to lend	Multiplier falls	Decreases	Contractionary
Reserve requirement down	Lending capacity rises	Multiplier rises	Increases	Expansionary
Open market purchase	Liquidity injected	Base increases	Increases	Expansionary
Open market sale	Liquidity absorbed	Base decreases	Decreases	Contractionary

Figure 3.1 The transmission map. Different mechanisms, symmetrical outcomes.

REAL-WORLD EXAMPLE

On 22 March 2020, as India entered lockdown, the RBI cut the repo rate by 75 basis points. Within forty-eight hours banks began reducing lending rates; within weeks the cost of borrowing fell for crores of people who had never once considered monetary policy.

EXAM POINTER

The high-value answer separates the tools by *what they act on*. Reserve requirements work on the multiplier; open market operations and the discount rate work on the base. Candidates who list three tools without that distinction lose the marks attached to the mechanism. For lender of last resort, the illiquid versus insolvent line is mandatory.

QUICK RECAP • TOPIC 3

- Three objectives in tension: growth, price stability, full employment.
- Three tools: open market operations, discount rate, reserve requirements.
- Base versus multiplier — reserve requirements change what can be multiplied, the others change what there is.
- Expansionary and contractionary policy use the same instruments in opposite directions.
- Lender of last resort supports the illiquid, not the insolvent — a bridge, not a bailout.

PART B • COMMERCIAL BANKING

Deposits in, credit out

4 What a Commercial Bank Actually Does

PART B • COMMERCIAL BANKING • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

A bank borrows short, lends long, and earns the spread — and every risk it carries flows from that single maturity mismatch.

Why this matters

Deposit ₹10,000 at a branch and by the time you walk out, part of it is heading to a farmer in Punjab and part to a founder in Bengaluru. Only a fraction stays in the vault. Understanding why is the whole of this topic.

Five core functions

Channel funds from surplus units to deficit units. **Support economic activity** by financing households, businesses and government. **Provide other services** — payments, remittances, wealth management, advisory. **Generate profit** through net interest margin and fee income. **Intermediate**, transforming risk, maturity and denomination for both sides.

All five rest on one identity: **total assets = total liabilities + capital**.

Asset transformation and the spread

Banks **borrow short** — deposits are often repayable on demand — and **lend long**, into home loans and project finance locked up for five, fifteen or thirty years. They charge more on loans than they pay on deposits, and that **interest spread** is the core of banking profitability.

Of every rupee deposited, the central bank mandates a fraction as *required reserves*. The remainder, *excess reserves*, is the lending fuel. Borrowing short to lend long creates the spread — and also the liquidity risk that must be managed.

The five Cs of credit

Before lending, banks screen the borrower. **Character**: track record and demonstrated willingness to honour obligations. **Capacity**: income and cash flow sufficient to service repayments. **Collateral**: assets pledged as security. **Conditions**: the macroeconomic environment, industry cycle and purpose of the loan. **Capital**: the borrower's own stake, which signals commitment and reduces moral hazard. All five must be evaluated together — weakness in any one can make an otherwise attractive loan unviable.

Five dimensions of bank management

Liquidity: hold enough liquid assets to meet withdrawals without distress. **Asset**: choose low-risk borrowers and diversify. **Liability**: raise funds at the lowest cost. **Capital adequacy**: hold equity to absorb losses, meeting the regulatory minimum under Basel III. **Risk**: monitor credit exposures and protect the net interest margin from rate movements. These are five lenses on one balance sheet, never managed in isolation.

BORROW SHORT, LEND LONG · THE SPREAD AND THE RISK BOTH COME FROM HERE
DEPOSITS IN

Short-term · repayable on demand ·
depositors want safety and liquidity

>

RESERVES HELD

A mandated fraction stays · the rest is
lending fuel

>

LOANS OUT

Long-term · home loans, infrastructure,
project finance

*The gap between the rate charged and the rate paid is the **interest spread**. The gap between the maturities is the **liquidity risk**.*

Figure 4.1 Asset transformation. One mechanism produces both the profit and the danger.

KEY IDEA

A bank is not a vault. It is a maturity-transformation machine, and its profit and its fragility are two consequences of the same design. Nothing else about banking makes sense until that is accepted.

EXAM POINTER

Name all five Cs — partial lists score proportionally — and give one line each. Where a question asks about bank risk, connect it explicitly to asset transformation rather than listing risks abstractly. The balance sheet identity is worth stating whenever functions are discussed.

QUICK RECAP · TOPIC 4

- Five functions: channel funds, support activity, other services, generate profit, intermediate.
- Total assets = total liabilities + capital.
- Asset transformation means borrowing short and lending long; the spread is the reward, liquidity risk the cost.
- Five Cs of credit: character, capacity, collateral, conditions, capital.
- Five management dimensions: liquidity, asset, liability, capital adequacy, risk.

5

Credit Creation and the Money Multiplier

PART B • COMMERCIAL BANKING • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

The banking system creates money by lending, and a single formula — one divided by the reserve ratio — tells you how much.

Why this matters

This is the most quantitative topic of the week and the one that closes the loop back to monetary policy. The reserve requirement in Topic 3 is the lever; this is the mechanism it operates.

Deposits do not sit idle

When you deposit money, the bank does not lock it in a vault. It becomes raw material. The bank holds back the mandated fraction as liquid reserves, lends the rest, and the borrower spends or transfers that loan — which then arrives as a fresh deposit at another bank, triggering another round. Each new deposit produces more reserves and more lending, and the money supply grows far beyond the original deposit.

Core idea: banks create money by lending. Every loan funded by a deposit triggers a new deposit elsewhere in the system.

The reserve requirement, and why it exists

The minimum percentage of every deposit a bank must retain as liquid reserves, set by the central bank and held as vault cash or on deposit at the central bank. It exists for three protective reasons: **liquidity protection**, so each bank can meet withdrawals on demand; **systemic stability**, preventing over-lending before credit risk cascades; and **prevention of bank runs**, breaking the chain of panic before contagion spreads.

The money multiplier

MM = 1 ÷ reserve ratio. For each ₹1 of fresh reserves entering the system, MM rupees of total deposits can be created. The relationship is inverse and exact: lower ratio, higher multiplier, more credit.

CREDIT CREATION AT A 10% RESERVE RATIO • ₹100 BECOMES ₹1,000

Bank	Deposit received	Reserves kept	Lent on	Cumulative deposits
A	₹100	₹10	₹90	₹100
B	₹90	₹9	₹81	₹190
C	₹81	₹8.10	₹72.90	₹271
D	₹72.90	₹7.29	₹65.61	₹344
E and beyond	Each round 10% smaller	...	...	Converging on ₹1,000

$MM = 1 \div 0.10 = 10$. No money was printed. The banking system created it by lending.

Figure 5.1 The credit creation chain. Each round is a tenth smaller than the last.

KEY IDEA

Nobody printed the extra ₹900. It exists because a promise to repay, backed by a fraction held in reserve, is itself spendable. That is why the reserve ratio is a monetary policy instrument and not merely a prudential rule.

REAL-WORLD EXAMPLE

Compare the outcomes at three ratios. At 5%, $MM = 20$ and ₹100 supports ₹2,000 of deposits — accommodative, used in a slowdown. At 10%, $MM = 10$ and ₹100 supports ₹1,000 — neutral. At 20%, $MM = 5$ and ₹100 supports only ₹500 — restrictive, used when an economy overheats.

EXAM POINTER

Numerical questions here are near-certain. State $MM = 1 \div \text{reserve ratio}$, compute the multiplier, then multiply by the initial deposit — and show the first three rounds of the chain if asked to explain the mechanism. Adding the inverse relationship in words secures the interpretation marks.

QUICK RECAP • TOPIC 5

- Banks create money by lending: each loan becomes a deposit elsewhere, repeating the cycle.
- The reserve requirement protects liquidity, ensures systemic stability, and prevents bank runs.
- $MM = 1 \div \text{reserve ratio}$.
- At a 10% ratio, a ₹100 deposit supports ₹1,000 of system-wide deposits.
- Ratio and multiplier move inversely, making the reserve requirement a policy lever.

6 Off-Balance Sheet Activity and Bank Risk

PART B • COMMERCIAL BANKING • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Banks now earn a substantial share of income from commitments that never appear on the balance sheet — which diversifies revenue and introduces a different class of risk.

Why this matters

Fee income has grown precisely because interest margins have thinned. The controls in this topic — value at risk and stress testing — return in Week 13 when derivatives desks use the same machinery.

What sits off the balance sheet

Commitments and contingencies that do not appear as assets or liabilities under normal conditions. They boost profitability through fee income without deploying the bank's own capital, diversify revenue away from thin interest margins, and serve multinational clients needing foreign exchange guarantees and hedging.

Three types. *Loan sales*: banks originate loans then sell them in the secondary market, earning fees and removing credit risk from the balance sheet. *Fee-based services*: foreign exchange, loan commitments, bank guarantees, letters of credit, standby credit lines. *Trading and risk management*: derivatives, futures, options, swaps and FX trading, used both to generate profit and to hedge exposures.

The controls

Value at risk estimates the maximum potential loss over a horizon under normal market conditions at a stated confidence level — for instance 99% over one day. **Stress testing** models extreme but plausible scenarios, identifying the tail losses VaR underestimates by design. **Internal controls** add position limits, approval hierarchies, deal-level audit trails and real-time monitoring. **Integrated oversight** has risk, finance and compliance review the books jointly, with escalation paths when limits are approached.

ON THE BALANCE SHEET, AND OFF IT

ON-BALANCE SHEET**RECORDED AS**

Assets and liabilities

REVENUE

The net interest margin spread

PRIMARY RISK

Credit and liquidity risk; default

CAPITAL REQUIRED

Yes — regulatory ratios apply

CONTROLS

Capital adequacy ratios

OFF-BALANCE SHEET**RECORDED AS**

Contingent items only

REVENUE

Fees, trading gains, commissions

PRIMARY RISK

Market and operational risk

CAPITAL REQUIRED

Minimal or none upfront

CONTROLS

VaR limits, stress tests, position caps

On-balance sheet activity builds the interest income base. Off-balance sheet activity diversifies it. Together they define a modern bank's earnings power.

Figure 6.1 The two halves of a modern bank's income, and the different risks attached to each.

COMMON PITFALL

Reading “off-balance sheet” as “hidden” or improper. These are legitimate, disclosed activities. The risk is not concealment but that they consume little capital upfront while creating contingent obligations that can crystallise suddenly.

EXAM POINTER

Name all three types with an example each, then give two control mechanisms with what each is for — VaR for normal conditions, stress testing for the tail. A comparison question wants the contrast drawn on revenue source, risk type and capital requirement.

QUICK RECAP · TOPIC 6

- Off-balance sheet items are commitments and contingencies that do not appear as assets or liabilities.
- Three types: loan sales, fee-based services, and trading and risk management.
- They generate fee income without deploying the bank's own capital.
- VaR estimates loss under normal conditions; stress testing covers the tail it misses.
- The trade-off is diversified revenue against market and operational risk.

PART C • INVESTMENT BANKING

Capital raising, advice and the deal

7 What an Investment Bank Is

PART C • INVESTMENT BANKING • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Investment banks take no deposits and make no loans. They live on the deal side of finance and are paid in fees, not interest.

Why this matters

The contrast with commercial banking is the single most examinable comparison in Week 2, and the history explains why the two were ever separate at all.

Start with what it is not

Unlike the banks you use daily, investment banks take no deposits and issue no loans — savings accounts and home loans are not their business. They operate in the capital markets, moving billions in a single transaction, serving corporations raising capital, governments issuing debt, and institutional investors making large allocations.

The revenue model

Fee-based, not interest-based. They charge a percentage of every deal executed. The smaller the deal, the higher the percentage, but the absolute fee scales with size — so the largest transactions are the most prized. A typical large-deal fee of around 3% of value means roughly \$300 million on a \$10 billion issue.

Four core functions

Underwriting: helping companies raise capital by issuing stocks or bonds. **M&A advisory:** guiding companies buying, selling or merging, including valuation and negotiation. **Intermediary services:** connecting buyers and sellers and managing the transaction. **Wealth management:** advisory for ultra-wealthy individuals and large institutions.

How the separation arose, and dissolved

Investment banking grew from nineteenth-century merchant firms that financed government bonds and then private ventures — J.P. Morgan financed American railroad expansion. Between 1900 and 1929 commercial banks moved deep into securities markets, and reckless lending into stocks fed directly into the 1929 collapse.

The **Glass-Steagall Act of 1933** legally split the two: commercial banks were barred from securities markets, investment banks prohibited from taking deposits. A wave of pure-play investment banks emerged and defined global banking for sixty years. From the 1980s the walls eroded — Bank of America bought Merrill Lynch, J.P. Morgan absorbed Bear Stearns — and Glass-Steagall was repealed in 1999, setting the stage for the 2008 crisis.

TWO SPECIES, ONE WORD • FUNDAMENTALLY DIFFERENT BUSINESS MODELS

COMMERCIAL BANK**CUSTOMERS**

Retail depositors and SMEs

CORE SERVICE

Deposits, loans and payments

REVENUE

The interest spread on loans

RISK TYPE

Credit and liquidity risk

REGULATION

Heavy — capital ratios, deposit insurance

EXAMPLES

HDFC, SBI, Wells Fargo

INVESTMENT BANK**CUSTOMERS**

Corporations, governments, institutions

CORE SERVICE

Capital raising and M&A advisory

REVENUE

Fees, around 3% of large deal value

RISK TYPE

Market and execution risk

REGULATION

Lighter — securities regulators such as SEC and SEBI

EXAMPLES

Goldman Sachs, Morgan Stanley

Figure 7.1 The comparison to memorise. Different customers, revenue, risk and regulator.**KEY IDEA**

The deposit is the dividing line. A commercial bank funds itself with other people's money and is regulated heavily because of it. An investment bank does not, and is regulated as a securities firm instead.

EXAM POINTER

Comparison questions want at least five axes: customers, core service, revenue model, risk type and regulation. Add the Glass-Steagall line — what each was allowed and banned from doing — and note the 1999 repeal. Dates carry marks here: 1933 and 1999.

QUICK RECAP • TOPIC 7

- Investment banks take no deposits and issue no loans; they operate in the capital markets.
- Revenue is fee-based, around 3% of large deal value, not interest income.
- Four functions: underwriting, M&A advisory, intermediary services, wealth management.
- Glass-Steagall (1933) split commercial from investment banking to protect depositors.
- Post-1980s deregulation and the 1999 repeal produced today's universal banks.

8 Underwriting and the IPO Process

PART C • INVESTMENT BANKING • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

The underwriter buys the entire issue at a fixed price and resells it — which is why pricing, not paperwork, is the decision that matters.

Why this matters

Every listing you read about in the financial press has been through this process. The pricing tension here is also the reason underpricing is a real cost rather than a technicality.

What underwriting is

The bank **buys the issue** outright at a fixed, agreed price, taking direct financial responsibility, then **resells it** to the market — the spread between purchase and resale is its compensation. It also advises whether to issue equity or debt, each carrying different cost and control implications, and times the market entry.

The pricing tension

The issuer wants a high price, because it maximises capital raised. The bank wants a realistic price, because if the issue is overpriced investors will not buy — and the bank already owns it. Overprice and the bank absorbs the shortfall from its own balance sheet; underprice and the issuer loses potential capital, money left on the table at launch.

An **IPO** is harder to price than a *seasoned reissue*, because no prior market price exists. A seasoned issue anchors to the current trading price and adjusts slightly; an IPO relies on comparables, models and judgement.

Bringing an issue to market

Registration with the regulator — SEBI in India, the SEC in the US — with a prospectus detailing financials, risks and use of funds. **Credit rating** by an independent agency. **Legal approval**, with counsel verifying compliance and a trustee appointed. **Syndicate** formation, so the lead bank shares underwriting risk and distribution. **Exchange listing** for secondary trading afterwards.

Three subscription outcomes, and two alternatives

Fully subscribed: demand matches supply. **Oversubscribed:** demand exceeds supply and allocations are scaled back. **Undersubscribed:** securities go unsold and the bank absorbs the shortfall.

Two arrangements shift that risk. Under **best efforts**, the bank sells what it can without committing to buy, so the issuer bears volume risk and the bank earns a commission. Under a **private placement**, securities are sold directly to large institutions with fewer regulatory hurdles, since the public is not involved.



Figure 8.1 Three components of flotation cost. Only two of them appear on a bill.

REAL-WORLD EXAMPLE

Zomato's July 2021 listing on the NSE and BSE closed roughly 66% above its IPO price on day one. That gain to investors is capital the company did not raise. In hot markets underpricing intensifies: euphoria raises the day-one pop, and the issuer leaves more behind.

COMMON PITFALL

Treating a large day-one gain as evidence of a successful IPO. From the issuer's side it is the opposite — it measures how much capital was forgone. Whose perspective the question takes determines the correct answer.

EXAM POINTER

State all three flotation cost components and identify underpricing as the hidden one. For process questions, name the five pre-issue steps in order and both risk-shifting alternatives with who bears the risk in each.

QUICK RECAP · TOPIC 8

- Underwriting means buying the whole issue at a fixed price and reselling it; the spread is the fee.
- Overpricing hurts the bank; underpricing costs the issuer capital.
- An IPO is harder to price than a seasoned reissue because no prior market price exists.
- Five steps: registration, credit rating, legal approval, syndicate, listing.
- Flotation cost = direct costs + underwriting spread + underpricing.

9 Mergers, Acquisitions and the Deal Cycle

PART C • INVESTMENT BANKING • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Investment banks identify targets, run tender offers, arrange financing and advise both sides — and leverage makes the whole business intensely cyclical.

Why this matters

The closing topic of the week, and the one that connects investment banking back to the credit conditions set in Part A. When credit is cheap, deals happen; when it freezes, they stop.

Merger, acquisition, friendly, hostile

A **merger** combines two firms into a single new entity: both dissolve into a new corporate vehicle, typically with a new name and structure. An **acquisition** is one firm buying another, with ownership transferring to the acquirer; the target may survive as a subsidiary or be absorbed.

In a **friendly takeover** both management teams agree and the acquirer negotiates with the target's board. In a **hostile takeover** the target resists, so the acquirer bypasses management with a *tender offer* — a public bid to buy shares at a premium, made directly to shareholders.

Four roles the bank plays

Identify targets by screening the market for strategic fit and ownership willingness. **Manage tender offers**: pricing, communication, regulatory filings, shareholder outreach. **Arrange financing** by structuring the capital stack of debt, equity, bridge loans and junk bonds. **Advise both sides**, through separate teams with ethical walls to prevent conflicts of interest.

Selling a business: the five stages

Valuation using market comparables, financial models and synergy potential. **CIM shared** — a confidential information memorandum goes to qualified buyers, only after non-disclosure agreements are signed. **Letter of intent**, non-binding, signalling price range, terms and structure. **Due diligence**, typically twenty to forty days verifying financials, legal exposure, operations and contracts. **Definitive agreement**: the final binding contract, at which ownership transfers.

Junk bonds and the cycle

Junk bonds carry elevated coupons compensating for high default risk and are issued with minimal collateral. Their significance is transformative: leverage replaces balance sheet, letting a buyer acquire targets far larger than itself — which is what made the roughly \$45 billion TXU buyout structurally possible.

The business is therefore **cyclical**. In booms, cheap financing and high confidence drive deal volumes and valuations up. In downturns credit tightens and activity collapses. In 2008 credit froze, Lehman Brothers failed, and weak institutions were absorbed by stronger ones in forced, survival-driven acquisitions. Leverage magnifies the upside in booms and amplifies the destruction in busts.



Figure 9.1 The deal cycle. Only the final stage binds either party.

KEY IDEA

Junk bonds are not merely risky debt. They are the instrument that decouples the size of an acquisition from the size of the acquirer — which is why they changed what deals were possible at all.

EXAM POINTER

Distinguish merger from acquisition precisely: a merger creates a new entity, an acquisition transfers ownership to an existing one. Name the tender offer as the hostile mechanism and say why it works — it goes around the board, directly to shareholders. For the sale process, name all five stages in order.

QUICK RECAP • TOPIC 9

- A merger creates a new entity; an acquisition transfers ownership to the acquirer.
- Hostile takeovers bypass management through a tender offer made directly to shareholders.
- Banks identify targets, manage tender offers, arrange financing and advise both sides behind ethical walls.
- Five sale stages: valuation, CIM, letter of intent, due diligence, definitive agreement.
- Junk bonds let small buyers acquire large targets; leverage makes the business deeply cyclical.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain the functions and evolution of money and how its supply is measured	Topics 1 and 2 · Figures 1.1, 2.1
2	Describe monetary policy tools, base versus multiplier, and financial stability	Topic 3 · Figure 3.1
3	Explain commercial banking, credit creation and bank risk control	Topics 4, 5 and 6 · Figures 4.1, 5.1, 6.1
4	Distinguish investment banking and explain underwriting, IPOs and M&A	Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1

The module in six sentences

Money exists to solve the double coincidence of wants, and performs three functions that no barter system can. Its supply is measured in graded layers from M1 to M3, and the central bank controls it with three tools — two acting on the monetary base, one on the multiplier applied to it. Commercial banks profit by transforming short deposits into long loans, earning the interest spread and carrying the liquidity risk that comes with the same mechanism. Credit creation multiplies deposits across the system, so that at a 10% reserve ratio a ₹100 deposit supports ₹1,000 of deposits, which is why the reserve requirement is a policy lever and not merely a prudential rule. Investment banks do none of this: they take no deposits, earn fees rather than interest, and buy entire securities issues at a fixed price, which makes pricing their central risk. Their history runs from nineteenth-century merchant financiers through the Glass-Steagall separation of 1933 to its repeal in 1999, and their business remains deeply cyclical because leverage makes it so.

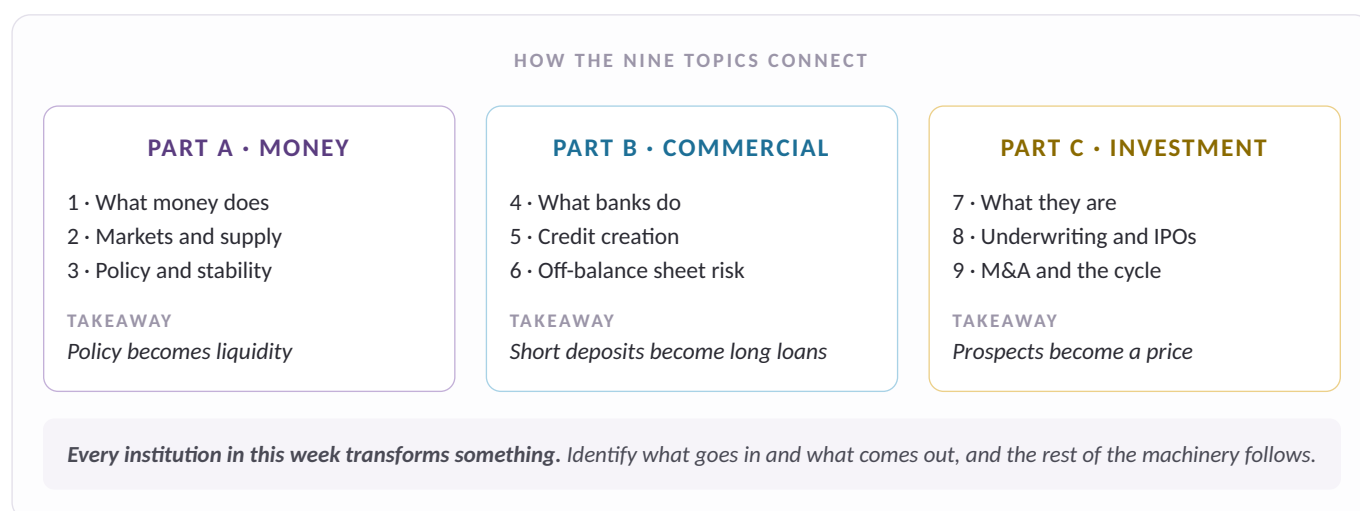


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Double coincidence of wants	The barter problem where two parties must each want exactly what the other has.
Fiat money	Currency declared legal tender by government, requiring trust in the issuing authority.
M1, M2, M3	Graded measures of money supply, each adding a less liquid component to the one before.
Velocity of money	The rate at which money circulates through the economy.
Monetary base	Currency in circulation plus all bank reserves held at the central bank.
Money multiplier	How far the banking system expands the base through lending. $MM = 1 \div \text{reserve ratio}$.
Reserve requirement	The minimum fraction of every deposit a bank must retain as liquid reserves.
Excess reserves	The portion of a deposit above the required reserve — the lending fuel.
Open market operations	Central bank purchase or sale of government securities to inject or drain reserves.
Discount rate	The rate at which banks borrow from the central bank.
Repo rate	India's policy rate, signalling the cost of money to the banking system.
Inflation targeting	A public commitment to a precise inflation target — the RBI's is 4%, within a 2-6% band.
Lender of last resort	Emergency central bank liquidity to institutions that are illiquid but not insolvent.
Moral hazard	The incentive to take reckless risk when rescue is expected.
Asset transformation	Borrowing short and lending long, converting illiquid risky assets into safer liquid ones.
Net interest margin	The spread between interest earned on loans and interest paid on deposits.
Five Cs of credit	Character, capacity, collateral, conditions and capital — the loan screening framework.
Capital adequacy ratio	The regulatory minimum equity a bank must hold to absorb losses, under Basel III.
Off-balance sheet activity	Commitments and contingencies generating fee income without appearing as assets or liabilities.
Value at risk	Maximum potential loss over a horizon under normal conditions at a given confidence level.
Underwriting	A bank buying an entire securities issue at a fixed price and reselling it to investors.
Underpricing	Setting an issue price below what the market would bear — capital the issuer forgoes.
Best efforts	An arrangement where the bank sells what it can without committing to buy the issue.

Term	Definition
Tender offer	A public bid to buy shares at a premium, made directly to shareholders over management's objection.
Junk bonds	High-yield debt with minimal collateral, letting buyers acquire targets larger than themselves.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A bank faces a 20% reserve requirement. A ₹100 deposit enters the system. How much total deposit can it support, and what happens if the ratio falls to 5%?

MM = $1 \div 0.20 = 5$, so ₹100 supports ₹500. At a 5% ratio, MM = 20 and the same ₹100 supports ₹2,000. Ratio and multiplier move inversely — which is why lowering the requirement is an expansionary act.

Q2 The central bank raises the reserve requirement and simultaneously buys government securities. What is happening?

Two tools pulling in opposite directions. Raising the requirement cuts the multiplier (contractionary); buying securities injects reserves and expands the base (expansionary). The net effect depends on magnitude — but the point is that one acts on the multiplier and the other on the base.

Q3 Why is the illiquid-versus-insolvent distinction central to the lender-of-last-resort role?

Because it separates a bridge from a bailout. An illiquid institution is solvent but cannot meet immediate obligations; lending to it prevents a contained run from spreading. Lending to an insolvent one merely postpones failure and rewards reckless risk-taking — moral hazard.

Q4 An IPO closes 66% above its issue price on day one. Is that a success?

It depends whose side you are on. For allotted investors, yes. For the issuing company it measures underpricing — capital it could have raised and did not. Underpricing is a real component of flotation cost, alongside direct costs and the underwriting spread.

Q5 Name three ways a commercial bank and an investment bank differ, beyond size.

Customers, revenue and risk. Commercial banks serve retail depositors and SMEs, earn the interest spread, and carry credit and liquidity risk. Investment banks serve corporations and institutions, earn fees of around 3% of deal value, and carry market and execution risk. Regulation differs too — capital ratios and deposit insurance against securities regulation.

Q6 Why does a bank's profit and its fragility come from the same mechanism?

Asset transformation. Borrowing short and lending long creates the interest spread that funds the bank, and simultaneously creates the maturity mismatch that makes it vulnerable to a sudden demand for withdrawals.